UNIFIED MECHANICS

The Whole Thing, Once

every result, every refusal, every grade, compressed to the point where nothing is left out and nothing is said twice

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WHY THIS DOCUMENT IS SHORT

The theory is invariant, so the document can be. Every fixed point is stated once, at full strength, with its grade on its face, and then referenced. Nothing is summarised and nothing is omitted: a summary loses content, and this loses none.

It is one object. The logic work and the physics work were built years apart in different vocabularies, and the joins were invisible because the terms changed underneath. They are the same structure. Where the old work names an open problem and the new work answers it in different words, both appear here in one line.

Grades are never mixed. **PROVED** follows from what precedes it. **CONDITIONAL** follows once a named import is granted. **PROPOSED** is a physical reading offered for falsification. **POSTULATE** is a starting commitment. **OPEN** is a question left standing, with the reason. **REFUTED** and **DISSOLVED** are things that were tried and did not hold, kept on the record because a corpus is worth most when it says what failed.

PART ONE

The question, and the one root

Take a string. Cut it into a longer piece and a shorter piece. Ask what ratio makes the whole relate to the longer exactly as the longer relates to the shorter. **That question contains no measurement.** There is nothing in it to fit to, so what comes out of it cannot be a fit.

$$\text{whole} : \text{longer} \;=\; \text{longer} : \text{shorter} \qquad \Longrightarrow \qquad \varphi^2 = \varphi + 1$$

$$\varphi = (1+\sqrt{5})/2 = 1.6180339887\ldots \qquad r = 1/(2\varphi) = 0.3090169944\ldots$$

PROVED. One equation, one positive root. The alternative method, asking which equation fits an observation, has infinitely many equally valid answers: every expression summing to twelve describes twelve cups. Six free parameters means six ways to be uninformative.

Squaring the two-pole unity is the only way two poles make a partition, and a square of two terms has exactly three terms. **Three is counted, not chosen.**

$$(u + r)^2 = u^2 + 2ur + r^2 = W_L + W_B + W_M = 1 \qquad u = 1 - r$$

	Value	Exact identity
W_L light	0.477457514	u^2
W_B boundary	0.427050983	$2ur$
W_M matter	0.095491503	r^2
W_L / W_M	5, exactly	since $u/r = \sqrt{5}$
$W_M W_L - (W_B/2)^2$	0, exactly	the geometric-mean identity
ε_{fl}oor	0.029508497	r^3, the price of one completed traversal

PART TWO

The reduction system, and its own audit

R1 to R5 on configurations and reductions. Everything downstream is a property of that and nothing else.

A configuration reduces; reductions branch and rejoin (R1, local confluence); every chain terminates (R2); a representation group acts equivariantly (R3); admissibility is decidable (R4); branching is bounded (R5, adopted as repair). The **frozen footprint** κ(x) is the unique normal form. The **trail** is the family of reduction paths from x to κ(x).

The framework was then audited against itself, exhaustively over all 558 admissible systems on $n \le 5$. **The casualties are the load-bearing part.**

	Statement	Grade
T1	Footprint equivariance, $\kappa \circ \sigma = \sigma \circ \kappa$	PROVED
T2	The two quotients commute; the square is a pushout	PROVED
T3	$x \approx y \Leftrightarrow$ equal footprint and equal trail volume	PROVED
T4	Conjecture 1.2 is a theorem at its stated scope, no postulate	PROVED
T5	The path-count measure is definable from composition alone	PROVED
T6	Refinement limits exist at infinitely many algebraic ratios	PROVED
TP	Trail persistence: the invisible history fixes visible statistics	PROVED
ML	Q(**E**) is the coarsest quotient carrying the canonical measure	PROVED
C1	Definition 9.2 separates a state from its own reduct, refuting its own commentary	REFUTED
C2	Time-grading is not derivable from R1 to R4	REFUTED
C3	Refinement limits over R1 to R4 exist at infinitely many ratios, so **the reduction axioms do not supply a second, independent route to r.** They were never the route: r is forced at the top, by the question, and this records that there is one road rather than two.	RESCOPED
R1′	Outcome equivalence 9.2′ restores the bridge; the observables fragment of Postulate A becomes a theorem	PROVED
R2′	Bounded degree as axiom R5: depth grading exists, four debts paid by one axiom	PROVED

THE TWO THINGS THIS AUDIT PRODUCED THAT NOTHING ELSE COULD

C3 asks the reduction axioms to do a job that belongs to the question, and reading it as a verdict on r is an error. r is forced in Part one: the string constraint gives one equation with one positive root, and $r = 1/(2\phi)$ normalises it. What C3 establishes is that starting instead from R1 to R4 and asking which branching ratio emerges admits infinitely many. That is a statement about there being **one** road to r, not about there being none. The four functionals tested were looking for a second road and did not find one; nothing required them to.

C1 converted a definitional slip into a physical fork. Q(**E**) retains footprint and trail volume; volume is not execution-invariant. So either execution volume is physical, and observables see computational volume, or it is not, and one adopts 9.2′ where everything heals. Both readings are consistent, mutually exclusive, and empirically distinct. **ML then broke the tie one way:** under 9.2 the canonical measure is a construction on the quotient rather than external bookkeeping, so that tine carries a universal property.

PART THREE

Three no-gos, each with exactly one escape

The framework does not advance by adding structure. It advances by closing doors until one remains open, which is the move the string question already made.

	What is forbidden	The single escape	Grade
Flatness F1	Loop-borne content. $b_1 = 0$ on every orbit complex computed; every loop bounds; holonomy trivial. **Because R1 fills every branching, the axiom itself forbids closed strings.**	Weaken R1 to coherent confluence: branches rejoin up to specified 2-cells. Curvature is the failure of local commutation, so a confluent system is flat by construction.	PROVED
Meyer D1	Scalar local unitary dynamics on the trail. Unitarity forces $a\bar{c} = 0$ and $a\bar{b} + b\bar{c} = 0$; only pure shifts survive.	An internal degree of freedom of dimension ≥ 2. The minimal choice is a chiral pair, and the massless case is the **exact** d'Alembert split with no action functional anywhere.	PROVED
LM-1a	Classical reversible dynamics. A stochastic matrix with stochastic inverse is a permutation matrix, so reversible classical evolution is a discrete set admitting no continuous flow.	The 2-norm, uniquely. For $p \neq 2$ the isometries are signed permutations; only $p = 2$ has a continuous isometry group. **Amplitudes and unitarity are the only continuous reversible dynamics available.**	PROVED

Those three compose into one chain, and its antecedent is a sentence that existed informally before it had a formal role: **nothing is ever discarded.** Written as reversibility, it is Trail Conservation.

Trail Conservation ⟹ not classical (LM-1a) ⟹ the 2-norm (LM-1b) ⟹ amplitudes

amplitudes + locality ⟹ chirality (D1, D2) ⟹ decohere ⟹ the canonical trail measure

PROVED as a chain; POSTULATE at the antecedent. D3 closes the loop downward: fully dephasing the forced channels returns the framework's own counting measure to an L1 error of 1.8×10^{-16}, with diffusive $\sigma = \sqrt{T}$ replacing ballistic $\sigma \propto T$. The classical statistics derived from pure path-counting is exactly the decoherent limit of the dynamics unitarity forces on the same arena. That is a consistency check the framework could have failed.

PART FOUR

What the trail recovers, and what is imported

The string mathematics the finite substrate reproduces, with every import named.

Item	Result	Grade
Oscillator tower	$\omega_n = 2\sin(\pi n/N)$ on the closed trail, to 10^{-8}	RECOVERED, exact
The −1/12	$\mathbf{E(N) = \Sigma \sin(\pi n/N) = \cot(\pi/2N) = (2/\pi)N - (\pi/6)/N + O(N^{-3})}$. The universal coefficient is $-\pi/6$ extracted to eight digits. On a finite substrate the notorious divergent sum is the subleading term of a convergent expansion. **No zeta function, no regulator, anywhere.**	RECOVERED
Central charge	$c = 1.0000000278$ from the Casimir coefficient	RECOVERED
Critical dimension	$a = (D-2)/24$ with the lattice $1/24$ gives $D = 26$	CONDITIONAL on the imported Virasoro intercept
Level degeneracies	Partition numbers $p(M)$ by explicit enumeration through $M = 12$: 1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, 56, 77	RECOVERED, exact
Hagedorn growth	$\log p(M) \to \pi\sqrt{(2M/3)}$	RECOVERED, asymptotic
Gravity sector	**576 = 299 + 276 + 1** under SO(24) by explicit orthogonal idempotents: one massless spin-2, one 2-form, one scalar. Right multiplicities and nothing extra.	PROVED
General relativity at low energy	Follows from the spectrum above via Weinberg's soft-graviton theorem	CONDITIONAL on that import plus LM-2 and LM-3
Level matching	On the closed trail, invariance under loop rotation forces $N = \bar{N}$ exactly. A consistency condition of string theory is a symmetry on this substrate, not a rule.	PROVED, truncation
Interactions	Reduction branching is a trail splitting and R1 is a joining vertex, so interaction vertices are confluence events. The only UM-native proposal in the chain.	PROPOSED
The tachyon	The level-(0,0) state has $M^2 < 0$. The bosonic closed string is unstable and the standard cure imports fermions and supersymmetry, neither of which anything in the framework produces. **Reported, not hidden.**	OPEN

THE NUMEROLOGY FIREWALL

$D - 2 = 24$ invites comparison with the exponent 240 and with every golden quantity in the corpus. **No such connection is computed, derived, or suggested, and none should be asserted without a theorem.** The temptation is recorded here so that it can be refused on the record.

PART FIVE

The fields

The same structure, read as physics. One assumption is admitted: the vacuum energy of space.

A least-action cell carries one quantum of action and is causal. Given the measured vacuum energy, its size is fixed and it lands on an identity that makes two hard problems into one missing number.

$$\varepsilon\tau = \hbar \qquad a = c\tau \qquad \rho_\Lambda = \varepsilon_0 / a_0^3 \qquad \Longrightarrow \qquad a_0 = (\hbar c/\rho_\Lambda)^{1/4}$$

Quantity	Value	Reading
a_0	88.11 μm	the whole cell
$a_0 \cdot W_M^{1/4}$	48.98 μm	its matter fraction, the sub-millimetre falsifier
$a_0^2 / (l_P L_\Lambda)$	**1.000000000000000**	**exact.** Newton's constant and the cosmological constant problem are the same missing number
a_0 / l_P	5.4517×10^{30}	the one ratio, appearing twice

Each cell splits into light, boundary and matter in the three weights, so each carries its own light barrier and its own matter scale. Four field scalars follow: α_L, α_M, ν and θ. **The map from those four to the three geometric potentials has rank three and nullity one**, kernel $\alpha_L \to \alpha_L + \delta$, $\nu \to \nu - 3\delta$.

PROVED, and reached twice independently. General relativity fixes both metric potentials and can never separate the light mode from the density mode, at any precision. The slip parameter comes out identically 1 for every response ratio, which is the second route to the same statement.

The colour field is the instrument. Every quantity carries a channel signature, so it carries a hue; the signature's total degree is a depth D; and what survives D crossings at the floor is a brightness. **The depth is read off the structure, so the tolerance band is fixed before any measurement is consulted.**

$$\text{hue} = \text{channel address} \qquad D = \text{crossings} \qquad \text{band} = 1 - (1-r^3)^D$$

PROVED. This replaces a tolerance rule of n floors with n chosen per observable, which is unfalsifiable in the direction that matters. Under the structural rule every measured entry in the suite falls inside its band, and the band was not chosen.

$\Lambda = 3/\sqrt{5}$ arrives twice by unrelated routes: once from MacDowell-Mansouri, and once as the composition bound of the colour field, where $W_B/(W_L+W_M) = 2ur/(u^2+r^2) = \sqrt{5}/3 = 1/\Lambda$ exactly. The bound $\arcsin(1/\Lambda) = 48.189685°$ then needs $360/48.189685 = 7.4705$ traversals to close, **selecting eight, the carrier dimension.**

> **WHAT Λ ACTUALLY IS: A MEAN RATIO, THREE AGAINST ONE**
>
> Written on the pole ratio $k = u/r$, the composition bound is $2k/(k^2+1)$, and setting it to $1/\Lambda$ gives $\mathbf{k + 1/k = 2\Lambda}$ identically. Inverting:
>
> $\mathbf{\Lambda = (k^2 + 1) / (2k)}$, which is the **arithmetic mean of k^2 and 1 over their geometric mean.**
>
> At $k^2 = W_L/W_M = 5$ this is $AM(5,1)/GM(5,1) = \mathbf{3/\sqrt{5}}$, exactly. The three is not an independent number: it is what 5 and 1 average to. The $\sqrt{5}$ is what they multiply to. **Three against one, and the five folds back to unity between them.**
>
> So Λ is a **consequence** of the partition, not an input to it, and the composition bound equalling 1/Λ is an identity rather than a selection. It corroborates the structure; it does not choose it. The choosing was done in Part one, by the question.

PROVED, symbolically. This also disposes of a circularity worry cleanly: there was no need for the gravity route to Λ to be independent of r, because Λ is not doing selection work. What it does is exhibit the same constant arriving from the mean structure of the partition and from MacDowell-Mansouri, which is corroboration between two derivations and still the harder kind of agreement to manufacture.

PART SIX

How a framework that does not sit still returns a number

The prediction is not the fixed point. It is the average over the evolution, and those differ whenever the operation is curved.

r is where a flow settles, not a postulate, and the flow does not rest there: exact r is unreachable in finitely many noisy cycles, and the recorded spread is the floor. So the number an observable returns is $\langle f(x) \rangle$ and not $f(\langle x \rangle)$, and the gap between them is the curvature of f against the spread. **Nothing in that is adjustable.**

$$x \to 1/(4x+2),\ \text{perturbed at } r^3 \qquad \langle f(x) \rangle \neq f(\langle x \rangle)$$

Observation	static f(r)	evolved ⟨f⟩	measured	moved
$\Omega_b = W_M/2$	0.047746 (−3.16%)	**0.048362 (−1.91%)**	0.049302	closer
$\Omega_D E$	0.688322 (+0.53%)	**0.686955 (+0.33%)**	0.684700	closer
$\Omega_c = W_B/\phi$	0.263932 (−0.21%)	**0.264683 (+0.07%)**	0.264500	closer
$Y_H e = W_L/2$	0.238729 (−2.56%)	**0.239000 (−2.45%)**	0.245	closer
n_s	0.963525 (−0.14%)	**0.964384 (−0.05%)**	0.964900	closer
$\tau = 2r^3$	0.059017 (+8.49%)	0.061109 (+12.33%)	0.054400	away

The one that moves away is the informative one. All three matter-channel entries are convex, so averaging raises all three; two were below the measurement and improved, and τ was already above it. **A mechanism that improved everything would be a free parameter in disguise.** τ then resolves on its own path: it is the only line-of-sight integral in the suite, and the closure count over that interval is already fixed at three by the Hubble basins.

$$\tau = 2r^3\,(1 - r^3)^3 = 0.0539451 \qquad \text{measured } 0.0544 \pm 0.0073 \qquad -0.06\sigma$$

PROVED as a reading. Three wins on assumption count, not on residual: it is the only option that imports no new number, because the three is already in the corpus doing other work. k = 0 contradicts what τ is; k = 1, 2 and 4 each require a free integer that nothing supplies.

PART SEVEN

The observable suite, in field vocabulary

The same dictionary the older programme carried as polynomials, read through the partition. Every line an exact identity to 25 digits.

	Field reading	Older form	Measured	Residual
Ω_b	**half the matter weight**, $W_M/2$	$r^2/2$	0.049302	−3.16%
Ω_c	**the boundary weight, one Born factor damped**, W_B/ϕ	$4r^2(1-r)$	0.264500	−0.21%
Y_He	**half the light weight**, $W_L/2$	$(1-r)^2/2$	0.245	−2.56%
N_eff	three channels plus one baryon share, $3 + W_M/2$	$3 + r^2/2$	3.044	+0.12%
n_s	unity less one matter weight at the leak rate, $1 - W_M(1-2r)$	$1 - r^2/\phi^2$	0.964900	−0.14%
A_s	seventeen crossings of the floor coordinate	r^{17}	2.100e-9	+1.74%
Ω_DE	what the partition has left	$1 - 9r^2/2 + 4r^3$	0.684700	+0.53%

Y_He/W_L = Ω_b/W_M = 1/2 is a symmetry the polynomial form hides completely. Read at the boundary rather than after it, both carry the un-charged traversal $1/(1-r^3)$, and both improve while the four entries that are **not** half a channel weight all move further away under the identical factor. The rule selects its own members.

$$\Omega_b = (W_M/2)/(1-r^3) = 0.0491975\ \ (-0.21\%) \qquad Y_He = (W_L/2)/(1-r^3) = 0.2459875\ \ (+0.40\%)$$

PART EIGHT

The sectors

Every closed form in r alone. The Standard Model carries nineteen free parameters across the same ground.

Sector	Closed form	Value	Measured	Residual
Fine structure	$1/\alpha = \phi^{10} + \phi^5 + \phi^2 + \phi^{-2}$	137.082	137.036	**+0.034%**
Weinberg angle	$\sin^2\theta_W = r/[2(1-r)] = 1/(2\sqrt{5})$	0.223607	0.22339	+0.32%
Proton to electron	$m_p/m_e = \phi^{15}(1+r)(1+r^3)$	1836.15	1836.153	+0.11%
Wolfenstein λ	$r/(1+r) = 1/\phi^3$	0.236068	0.224500	+5.15%
CP phase	$\delta_CP = \arccos(W_B)$	1.1296 rad	1.14 ± 0.13	−0.92%
b to c amplitude	$\lvert V_cb\rvert = W_B \cdot W_M$	0.040780	0.041130	−0.85%
Baryon asymmetry	$\eta_B = r^{18}$, three generations by six crossings	6.602e-10	6.104e-10	+8.2%
Muon hierarchy	$m_\mu/m_e = \phi^{11}(1+r^3)$	204.877	206.768	−0.92%
Tau hierarchy	$m_\tau/m_e = \phi^{17}(1-r^3)$	3465.63	3477.23	−0.33%
Higgs to Planck	$r^{33}(1-r)$	1.0207e-17	1.0259e-17	−0.51%
Effective G	$G_eff/G_N = 1 + 1/(2\phi^4)$	1.072949	—	—
Vacuum readout	$\lambda = (4\pi/\sqrt{3})\, r^{240}$	2.86033e-122	2.845e-122	**0.28σ**
Clustering	$D = 2 + 1/(2\sqrt{5})$	2.223607	2.0 to 2.2	in range

TWO CONVENTIONS THAT ARE NOT INTERCHANGEABLE

The vacuum sector carries both $\mathbf{r^{240}}$ and $\mathbf{r^{241}}$, and they are numerically distinct. The completed readout $(4\pi/\sqrt{3})r^{240}$ sits 0.28σ from the measured cosmological constant. The bare core r^{240} is a factor 7.26 away, and comparing it in log-magnitude hides that: a 0.4 per cent agreement in the exponent is a 620 per cent disagreement in the number. **The boundary completion is not decoration, it is the whole match.**

The depth reading $r^{241} = 1.218 \times 10^{-123}$ against $\rho_\Lambda/\rho_P l = 1.155 \times 10^{-123}$ is a different normalisation of a different quantity, at +5.5 per cent, two floors. Both are on the record; neither may be substituted for the other.

PART NINE

The two basins

Seven invariants and one temperature anchor, integrated forward. No Boltzmann hierarchy, no fitted parameter.

Recombination by Saha, sound horizon and comoving distance by forward integration, and the scale solved from the acoustic angle rather than supplied. **The two Hubble measurements are two basins, not two estimates of one number**, and the closure count is the ratio between them. Anchoring on a light-channel observable returns the laminar basin; the distance ladder is measured in the condensed flow.

Basin	H_0	Against	Pull
laminar, θ_* anchored	**66.97**	Planck 67.36 ± 0.54	-0.73σ
condensed, $\times (1-r^3)^{-3}$	**73.26**	SH0ES 73.04 ± 1.04	$+0.21\sigma$
condensed, $\times (1+3r^3)$	72.89	SH0ES	-0.14σ
condensed, symmetric form	73.17	SH0ES	$+0.12\sigma$

All three closure conventions land inside a quarter of a sigma, and they differ from each other by less than the floor, so the sector does not need to choose between them. Reading Ω_b at the boundary takes the acoustic angle to **−0.014%**, ω_b to **−0.13%**, and the laminar basin to **67.39, which is $+0.05\sigma$ from Planck.**

INDEPENDENT EVIDENCE, ARRIVING IN THE RIGHT ORDER

The basin structure was not predicted and then looked for. MCMC chains weighted toward matter-clustering likelihoods passed every standard convergence diagnostic, $\hat{R} < 1.01$ with stationary traces, and returned χ^2 excesses of **+1414 and +1460** against the reference. They had converged correctly to the wrong basin, at $h \approx 0.7362$.

The weighting that avoids the trap is roughly 70 / 25 / 5, light to boundary to matter, **which is the channel partition.** It was found by chain failure and explained afterwards. Evidence that arrives in that order carries far more weight than a formula that matches after the fact.

PART TEN

Mass, and where it was not

A line the corpus carried open for years, closed by finding that the search was in the wrong place.

No element of so(3) + so(4,1) has three distinct nonzero real eigenvalues on the boundary block. Two hundred random elements reach a maximum of two, and the reason is structural: family weights are purely imaginary, so only zero-family-weight states carry a real part. **Attacks on the mass problem were looking for three masses in a place that cannot hold them.**

The operator is in the boundary block, and it is $\mathrm{ad}(X)^2$ rather than $\mathrm{ad}(X)$ because $[15,15] \subset 3 + 10$, so $\mathrm{ad}(X)$ has no fifteen-to-fifteen component at all. That is also right physically, since an action carries mass squared. On the three (family, time) directions:

$$m^2 = v_1^{\,2} + v_2^{\,2} + v_3^{\,2}$$

PROVED. One eigenvalue in a Pythagorean relation. ***The families are components of the operator, not eigenvalues of it****, which is why no spectrum ever showed three. Twelve components remain and they carry the splitting, fixed by stationarity under the same action that gave $B_{15} = -R_{15}$. At this order the three families come out degenerate, which is wrong, and the Yukawa hierarchy is the standing problem. What changed is that its location is known rather than searched for.*

PART ELEVEN

The instrument

Resolution belongs to the object. What a thing's own spread says about it, before anyone measures.

Define resolution as a quantity over its own spread, $N = q/\delta q$. Read δq as measurement error and it describes the observer; read it as the quantity's own spread and it describes the object. An oblate rotating body genuinely has different gravity at equator and pole, so g has no single value to know.

$$H_5(N) = 2s^{10}/(1+s^{10}) = 1 + \tanh(5 \ln s_N) \qquad s_N = N/(N+1) \qquad \text{deficit} = 5/N$$

PROVED. The five is the observer-state dimension: three of space, one of passage, one of record, because two histories can share an endpoint and a clock and still differ. So the exponent was never free, and $s^{10} = (s^5)^2$ is one traversal out and back. The general member is H_d with deficit d/N, and Fermi-Dirac is the $d = 1$ case: exclusion makes occupancy binary, and two branches compared and normalised has only one possible shape. Temperature is the unit the logarithm is expressed in, not structure.

Body	N	Deficit	Compactness	Side of $40\phi^3$
Saturn	5.40	69.0%	1.47e-08	below
Jupiter	8.83	49.0%	4.08e-08	below
PSR B1937+21	**10.59**	42.3%	**3.46e-01**	below
Neptune	32.86	14.9%	6.20e-09	below
Mars	121.82	4.09%	2.82e-10	below
Earth	196.38	2.54%	1.39e-09	above
Sirius B	**9666**	0.052%	**5.28e-04**	above
Sun	93534	0.005%	4.25e-06	above

The reading is decoupled from gravitational strength. Sirius B carries two thousand times Earth's compactness and is among the sharpest bodies in the table. A millisecond pulsar is the most compact entry by seven orders of magnitude and sits below the join beside Jupiter. Order the table by compactness and by N and the orderings are unrelated: N tracks flattening and spin. **An object's internal state is not recoverable from its external one.**

PART TWELVE

The ledger, and the refusals

What stands, what is owed, and what was tried and let go. A corpus is worth most where it says what failed.

The dependency graph carries 71 nodes and 82 dependency edges, acyclic, every edge naming the phrase that licenses it. **No edge runs from an empirical node into a structural one, and the single numerical node is depended on by nothing.** Twenty-three foundations depend on nothing: six axioms and sixteen definitions. For any result, walk down to an axiom without passing through a measurement.

Standing	Item
RESCOPED	The extremal-family problem asked the reduction axioms for a second, independent derivation of r. There is one road and it is the question, Part one. Four stability functionals confirmed no second road exists over R1 to R4, which costs the framework nothing.
OPEN	Trail Conservation from R1 to R5, rather than as a postulate. The framework's entire quantum content hangs from it.
OPEN	The tachyon. Removing the level-zero instability needs worldsheet fermions, which nothing in the framework produces.
OPEN	The twelve splitting components of the mass operator, and with them the Yukawa hierarchy.
OPEN	The microscopic matter-to-cell law. The one step between representing general relativity and deriving it.
OPEN	Emergent Lorentz invariance on the target space, beyond the worldsheet. The causal order natively carries conformal symmetry, which is the same object as the one import in the $D = 26$ chain, approached from the other side.
DISSOLVED	The action lock $T·P = 1$. Asserted, not derived: $\varepsilon\tau = \hbar$ is constant by definition and the cell area is not the action.
DISSOLVED	Schwarzschild derived with no field equation. The profile was supplied to the integrator. What stands is the exact representation.
DISSOLVED	The two-body force calculation. Started from the Newtonian potential and recovered Newton. A change of variables.
DISSOLVED	The QCD string tension. $\sigma = \rho a^2 = E^2/(\hbar c)$ identically, so supplying a scale and recovering the tension is a unit conversion.
DISSOLVED	Crowding energy as dark matter. Short by 10^{11} in galaxies, and structurally bounded by the vacuum energy.
DISSOLVED	The charged-lepton mass formulas as predictions. The muon candidate misses by 0.92 per cent against a quantity measured to two parts in 10^8.
DISSOLVED	The electroweak VEV prefactor. Three chosen factors for 0.1 per cent. The geometric-mean bridge survives; the prefactor does not.
DISSOLVED	Two apparent contacts between H_5 and the corpus constants. Exact, and carrying no information: for any r, $s_N = r$ at $N = r/(1-r)$, so the content belongs to the golden ratio and not to the operator.
DISSOLVED	The maximum-leverage resolution as 2π. It is 6.2825413 against 6.2831853, one part in 10^4, and a smooth maximum landing that close is coincidence. Logged so the week is not spent.

WHAT WOULD END IT

A confirmed gauge structure outside so(10). A fourth light generation. Dark matter carrying visible gauge charge. An observer boundary of nonzero genus. **Sub-millimetre gravity showing no cell structure near 48.98 microns**, which is just under the limit torsion balances have already reached. And the standing one, which has not been run: a running, self-holding system that fails to return quantised retention at powers of r, under a convention declared before the count.

One rule governs every map. **No free parameters, because a knob can manufacture any agreement and therefore certifies none.**

PROVENANCE

Assembled from the corpus, the older repository, the Supporting theorem notes, the nine sector papers and addenda A01 to A05. Every number here regenerates from a script in one of those places. The Monograph stands by itself and tells the story of how everything forms; nothing here replaces a line of it. This document exists because the logic work and the physics work were built in different vocabularies and nobody had written the dictionary between them.